

An Audited Baseline and Proof Interfaces for Erdős Problem 892

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22 August 2026

Abstract

We give a conservative research baseline for the two questions known as Erdős Problem 892. The purpose is not to claim a solution, but to separate elementary and published consequences from an extensive supplied partial-results manuscript whose new claims still require independent audit. We prove the exact counting and compactness interfaces for domination, verify the historical meaning of the gcd-free special case, derive the harmonic obstruction for prescribed density scales, and state four precise remaining proof targets. We also specify a branch and formalization protocol under which only gap-free, independently reviewed results may enter the main proof.

1 The problem and its historical convention

A set $A \subseteq \mathbb{N}$ is *primitive* if no two distinct members divide one another. Write $A(x) = |A \cap [1, x]|$.

Question 1.1 (Domination). For which strictly increasing integer sequences $b_1 < b_2 < \dots$ do there exist a primitive sequence $a_1 < a_2 < \dots$ and $C < \infty$ such that

$$a_n \leq Cb_n \quad (n \geq 1)?$$

Question 1.2 (Prescribed density scales). For which strictly increasing exponent sequences $n_1 < n_2 < \dots$ do there exist a primitive set A and $c > 0$ such that

$$A(2^{n_i}) \geq c2^{n_i} \quad (i \geq 1)?$$

The original conference paper states the special hypothesis $\gcd(b_i, b_j) \neq b_r$ only for incomparable b_i, b_j [4, pp. 41–42]. We therefore make the following definition.

Definition 1.3. A set $B \subseteq \mathbb{N}$ is *quasi-primitive* if

$$x, y, d \in B, \quad x \nmid y, \quad y \nmid x \implies \gcd(x, y) \neq d.$$

If the phrase “no non-trivial solution” were instead read as excluding every gcd identity except $i = j = r$, then $b_i \mid b_j$ for $i \neq j$ would already be impossible, so B itself would be primitive. That literal reading makes the special case trivial and is not the historical question.

2 Exact domination interfaces

Let $B(x) = |\{n : b_n \leq x\}|$.

Proposition 2.1 (Counting formulation). *For an increasing sequence $A = \{a_1 < a_2 < \dots\}$ and fixed $C \geq 1$, the following are equivalent:*

- (a) $a_n \leq Cb_n$ for every n ;
- (b) $A(Cx) \geq B(x)$ for every real $x \geq 1$;
- (c) $A(Cb_n) \geq n$ for every n .

Proof. If $B(x) = m$, then $b_m \leq x$, hence $a_m \leq Cb_m \leq Cx$ and $A(Cx) \geq m$. Thus (a) implies (b), while (b) implies (c) by taking $x = b_n$. Finally (c) says that the n th element of A is at most Cb_n , proving (a). $\square$

Proposition 2.2 (Finite-prefix compactness). *Fix an integer $C \geq 1$. There exists an infinite primitive sequence with $a_n \leq Cb_n$ for all n if and only if, for every N , there is a primitive tuple*

$$c_1 < \cdots < c_N, \quad c_i \leq Cb_i \quad (1 \leq i \leq N).$$

Proof. Necessity follows by truncation. Conversely, form a rooted tree whose level- N vertices are the feasible N -tuples, with edges given by extension. Every vertex has finitely many children because the next coordinate is at most Cb_{N+1} , and every level is nonempty by hypothesis. König's infinity lemma gives an infinite branch; its union is the required primitive sequence. $\square$

Proposition 2.2 is an exact necessary-and-sufficient condition but not yet an arithmetic characterization. For the quasi-primitive special case, the missing assertion is a uniform bound on the finite dilation constants, not merely the existence of a witness for each fixed prefix with a prefix-dependent constant.

3 A sufficient theorem for arbitrary irregularity

The following result appeared in the supplied manuscript and has now received two independent line-by-line audits. Its proof is included in full.

Theorem 3.1 (Dyadic reciprocal-load criterion). *Let $b_1 < b_2 < \cdots$ be positive integers and suppose*

$$H := \sum_{n=1}^{\infty} \frac{1}{b_n} < \infty.$$

If K is an integer with $K \geq \max\{4, 4H\}$, then there is a primitive sequence $a_1 < a_2 < \cdots$ satisfying

$$a_n \leq 2Kb_n \quad (n \geq 1).$$

Proof. For $j \geq 0$ put

$$D_j = \{n : 2^j \leq b_n < 2^{j+1}\}, \quad m_j = |D_j|, \quad I_j = (K2^j, 2K2^j] \cap \mathbb{N}.$$

Distinctness of the b_n gives $m_j \leq 2^j$, while I_j has exactly $K2^j$ members. We recursively choose m_j targets from I_j .

Suppose all earlier target blocks have been selected and let $E_{<j}$ be their union. Since a target chosen from I_i exceeds $K2^i$,

$$R_{<j} := \sum_{a \in E_{<j}} \frac{1}{a} < \frac{1}{K} \sum_{i < j} \frac{m_i}{2^i}.$$

For $n \in D_i$ we have $b_n < 2^{i+1}$, and consequently

$$\frac{m_i}{2^i} < 2 \sum_{n \in D_i} \frac{1}{b_n}.$$

Thus $R_{<j} < 2H/K \leq 1/2$. We also have

$$|E_{<j}| = \sum_{i < j} m_i \leq \sum_{i < j} 2^i = 2^j - 1.$$

Delete from I_j every multiple of an earlier target. A fixed $a \in E_{<j}$ has at most $K2^j/a + 1$ multiples in this interval, so the union bound deletes at most

$$K2^j R_{<j} + |E_{<j}| \leq \frac{K}{2}, 2^j + 2^j - 1.$$

At least

$$K2^j - \frac{K}{2}, 2^j - 2^j + 1 = \left(\frac{K}{2} - 1\right) 2^j + 1 \geq 2^j \geq m_j$$

candidates remain. Choose any m_j of them.

Each target block is primitive because two distinct integers in $(X, 2X]$ cannot divide one another. Every later block is strictly larger than every earlier block; size excludes divisibility from later to earlier, while the deletion rule excludes divisibility from earlier to later. The union is therefore primitive. Finally, the source bins and target blocks occur in the same order. Matching within each block increasingly, $n \in D_j$ implies

$$a_n \leq 2K2^j \leq 2Kb_n.$$

□

This criterion permits arbitrary burstiness. It is not necessary: a sequence of order $n \log n$ is dominated by the primes although its ordinary reciprocal sum diverges.

4 Published necessary conditions

Erdős proved that every primitive A satisfies

$$\sum_{\substack{a \in A \\ a > 1}} \frac{1}{a \log a} < \infty$$

[2]. Erdős–Sárközy–Szemerédi later proved that every infinite primitive A satisfies

$$\sum_{\substack{a \in A \\ a < x}} \frac{1}{a} = o\left(\frac{\log x}{\sqrt{\log \log x}}\right)$$

[3]. Applying these to a dominator and using $a_n \leq Cb_n$ gives the two classical necessary conditions displayed on the problem page:

$$\sum_n \frac{1}{b_n \log b_n} < \infty, \quad \sum_{b_n < x} \frac{1}{b_n} = o\left(\frac{\log x}{\sqrt{\log \log x}}\right).$$

The comparison is unaffected by finitely many terms or by replacing x with Cx .

Martin and Pomerance prove a substantial approximate converse for smoothly varying counting profiles [5]. It does not coordinate arbitrary abrupt bursts, which are central to both questions here.

5 A necessary condition for prescribed scales

Theorem 5.1. *If a primitive set A and a constant $c > 0$ satisfy $A(2^{n_i}) \geq c2^{n_i}$ for every i , then*

$$\sum_i \frac{1}{n_i} < \infty.$$

Proof. We may take $0 < c \leq 1$. Put $X_i = 2^{n_i}$ and $\delta = c/2$. For all sufficiently large i ,

$$|A \cap (\delta X_i, X_i]| \geq A(X_i) - \lfloor \delta X_i \rfloor \geq \frac{c}{3} X_i.$$

Every element of this slice contributes at least $(X_i \log X_i)^{-1}$ to the Erdős sum, so the slice contributes at least

$$\frac{c}{3 \log X_i} = \frac{c}{3(\log 2)n_i}.$$

Choose an integer r with $\delta 2^r > 1$. Since the n_i are strictly increasing integers, the slices with indices in any fixed residue class modulo r are pairwise disjoint. If $\sum_i 1/n_i$ diverged, one residue-class subseries would diverge and force the Erdős sum of A to diverge, a contradiction. $\square$

6 The supplied partial-results manuscript

The 169-page stage manuscript [1] explicitly states that it does not solve either characterization or the quasi-primitive special case. It proposes many strong partial theorems. The present project treats them as candidates rather than inherited facts until their hypotheses, analytic citations, and proofs are independently checked. The four terminal interfaces extracted from its dependency ledger are:

1. for quasi-primitive inputs, either a universal bounded-dilation embedding or a genuine divergent obstruction;
2. a necessary-and-sufficient cross-scale condition for arbitrary domination beyond scalar and consecutive-load restrictions;
3. a converse, or further obstruction, for exponent sequences satisfying harmonic, global, and uniform local crowding restrictions;
4. an actual-union estimate for middle-anchor low-multiplier events in a terminal short interval, where natural-density and raw second-moment methods are reported to fail.

The most concrete first-question finite target is the conjectural universal factor-two witness for every finite quasi-primitive tuple. Finite verification cannot by itself prove the infinite assertion, but a uniform theorem would combine immediately with Proposition 2.2. Conversely, a single finite tuple requiring a constant greater than two would refute only the factor-two strengthening, not the original bounded-constant question.

7 Proof dependency and admission criteria

Any complete solution must supply a chain from one of the exact compactness interfaces to a verifiable arithmetic condition. A route is not discarded for lack of a standard tool. It is closed

only by a counterexample, contradiction, or a theorem that its precise mechanism cannot imply the required node. Failed routes are retained as barrier theorems with their logical scope stated explicitly.

A final solution claim requires: (i) a complete paper proof; (ii) two independent adversarial reviews of every critical lemma; (iii) deterministic certificates for all finite computations; and (iv) a Lean theorem with no placeholders or project-defined axioms. None of these conditions is presently asserted to hold for the full problem.

8 Conclusion

The exact historical problem remains open in this repository. The contribution of this baseline is to fix the convention, prove the elementary reductions, pin the published necessary inputs, and expose the precise interfaces at which new mathematics is needed. Subsequent branch work will be promoted only when it strengthens this audited ledger without silently assuming one of the open interfaces.

References

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